Multilingual Voice Assistants for Intelligent Vehicles: A Survey of AI Techniques and Cross-Cultural Challenges

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Abstract

Multilingual voice assistants (VAs) are becoming integral to intelligent vehicle systems, enabling hands-free interaction across diverse linguistic and cultural contexts. This paper surveys the state-of-the-art AI techniques powering these systems, with emphasis on technological advancements and cross-cultural usability challenges. We examine and compare existing approaches based on evaluation aspects such as language coverage, accent and dialect robustness, real-time responsiveness, contextual understanding, and cultural adaptability.

Recent innovations in language-agnostic speech recognition and neural transducer-based speech translation have enabled robust multilingual processing across a wide range of languages [1, 2]. However, significant challenges remain in addressing regional speech variations, code-switching, and culturally embedded expressions, which can reduce effectiveness—particularly for older and multicultural user groups [3–5].

Moreover, integrating emotion-aware interaction models and multimodal sensor fusion is essential for delivering adaptive, user-centered experiences. These approaches foster safety, personalization, and trust through culturally sensitive interaction design [3]. This paper reviews and structures existing research to highlight key trends in speech AI, cross-lingual training, and explainable, privacy-aware systems, while identifying gaps in language inclusivity and deployment strategy. It aims to support the development of accessible, intelligent, and culturally responsive in-car voice assistants.

Keywords: Multilingual Voice Assistants, Intelligent Vehicles, Artificial Intelligence, Speech Recognition, Natural Language Processing, Cross-Cultural HMI, In-Vehicle Interaction.

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Chapter 1

Introduction

1.1 The Evolution of In-Vehicle Human-Machine Interaction (HMI)

Over the last few decades, how humans interact with vehicles has changed significantly. In-vehicle Human-Machine Interfaces (HMIs) have shifted from basic mechanical controls and simple analog displays to advanced digital systems. Early car HMIs featured physical knobs, buttons, and straightforward visual outputs, demanding a lot of the driver's manual and visual attention. The introduction of in-car entertainment in the mid-20th century, followed by navigation systems towards its end, marked early moves towards more complex digital interactions inside the car. However, the real game-changer for in-vehicle HMIs has been the rise of intelligent vehicles and Advanced Driver-Assistance Systems (ADAS) in the 21st century [6]. Modern vehicles frequently include large touchscreens, smartphone integration, and numerous connected services, offering drivers more information and control than ever before.

This boom in in-vehicle technology, while offering many advantages, has also made HMIs more complicated. Drivers now manage a wide range of information sources and control options, from detailed climate settings and media management to advanced navigation and ADAS configurations. This complexity can, surprisingly, increase a driver's mental workload and cause distractions, potentially affecting road safety [7]. As a result, the automotive industry and Human-Computer Interaction (HCI) researchers are actively seeking more intuitive, natural, and, importantly, hands-free ways for people to interact with their vehicles. Voice User Interfaces (VUIs), powered by voice assistants (VAs), stand out as a key solution. They promise to boost safety by minimizing manual and visual distractions and improve the user experience by providing a more natural way to control systems and access information [3, 5]. The ability to simply speak commands for navigation, music playback, or making calls—without taking hands off the wheel or eyes off the road—highlights the significant impact voice control can have in cars. This is especially relevant for connected and autonomous vehicles (CAVs), where the concept of "driving" itself is evolving, creating new opportunities for how occupants use their time and engage with the vehicle's intelligent systems [8].

1.2 Why Multilingual VAs are Needed in a Global Car Market

The automotive industry is a truly global business. Major manufacturers sell cars in many international markets, serving customers worldwide from diverse linguistic and cultural backgrounds. This global reach means in-car tech, especially HMIs, needs to support multiple languages to be accessible and sell well [9]. Users increasingly want to use technology in their own language because it's usually more comfortable, easier to understand, and provides a better experience [10, 11]. For many people, like those in areas with multiple languages or living abroad, using tech in a non-native language can make it hard to use effectively and enjoyably.

So, making VAs multilingual isn't just about convenience; it's key for including more users and reaching wider markets. To serve non-native speakers and diverse language groups, VAs need to

handle many languages and dialects. While English has historically been the main language for VA development, it's clear that global products need to move beyond just one language [9, 12]. Current in-car VAs often support only a few major languages or specific local versions, leaving out many users worldwide [10]. Even when they support several languages, how well they work can vary across them [13]. Also, many current VAs only handle one language at a time. This doesn't work well for bilingual or multilingual people who often mix languages (code-switch) when they talk [11, 14]. Fixing these issues is vital for creating smart car HMIs that are advanced, user-friendly, and available to everyone globally. For instance, the "Talk2Care" system, while designed for healthcare, highlights usability problems that can arise, even for older adults, if systems aren't intuitive—a lesson that also applies to the focused interaction needed when driving [15].

1.3 What are Multilingual Voice Assistants in Smart Cars?

A multilingual voice assistant for smart cars is an AI system that can understand and speak in several languages. This helps drivers and passengers control car functions, get information, and use communication services hands-free. Such a system uses a series of speech and language processing tools, all adjusted for multiple languages and the specific needs of a car environment.

First, the system needs strong **Multilingual Automatic Speech Recognition (ASR)**, or Speech-to-Text (STT). This part accurately converts spoken commands and questions from users in different languages, dialects, and accents into text, even with background car noise [16–18]. After ASR, **Multilingual Natural Language Understanding (NLU)** figures out what the transcribed text means. This means it identifies what the user wants to do (like "go to the airport," "play classical music," or "call mom") and pulls out key details (like the destination, music type, or contact name) from what was said, no matter the language [19, 20].

Once the system knows the user's intent, **Dialogue Management (DM)** manages the conversation, keeps track of the context over several turns (even if languages change), and decides how the system should respond. This is especially tricky when users switch or mix languages in one conversation (code-switching) [13, 14]. Lastly, **Multilingual Speech Synthesis (TTS)**, or Text-to-Speech, along with **Natural Language Generation (NLG)**, create a clear and natural-sounding spoken reply in the right language. This could be the language the user spoke in or another one they prefer [3, 12].

Good multilingual VAs in cars need more than just basic command-response; they need strong **contextual awareness**. This involves knowing the current driving situation (like speed and traffic), the car's status (like fuel level or ADAS info), user preferences learned over time, and even understanding common phrases and cultural references in the supported languages [4, 21]. Handling different accents well is very important, as studies show users from various cultures and non-native speakers get frustrated when VAs don't understand them [5, 11]. A multilingual VA in a car should skillfully manage main driving tasks like setting navigation and controlling ADAS, as well as other tasks like managing music and climate, making hands-free calls and messages, and giving emergency help or car diagnostic info [22–24].

1.4 Scope and Goals of this Research Summary

Quick progress in AI, especially in speech and language processing, along with a growing need for global and user-focused car technologies, has led to more research on multilingual voice assistants for smart cars. But this knowledge is often scattered across different areas like AI, HCI, and automotive engineering. This research summary aims to fix this by organizing, combining, and evaluating the current knowledge in this growing area.

The main **goals** of this survey are:

1. To give an overview of current AI techniques for multilingual VAs in smart cars (ASR, NLU, DM, MT, TTS).

2. To group and compare different tech approaches, noting their pros, cons, and how well they fit in cars (e.g., model types, training methods, performance like accuracy, speed, language support).
3. To spot and analyze key trends, like the growing use of Large Language Models (LLMs), language-neutral systems, and better handling of code-switching and accents.
4. To look closely at ongoing challenges, especially cross-cultural usability (like regional speech, cultural details, and needs of users like older adults or non-native speakers) [4, 11, 21].
5. To check how useful current research is and find gaps between lab results and real-world car use.
6. To combine existing literature to suggest ways to evaluate and best practices for developing future in-car VAs that are accessible, smart, and culturally aware.

It's important to note that this research summary will *not* present new findings or experiments. Instead, its main contribution is to carefully review, compare, and assess *existing work*. By doing this, we aim to check ideas found in the literature (like LLMs being better at context but possibly slow in cars) and see how well the discussed AI methods work in real life.

1.5 How This Survey Was Done

This research summary is based on a systematic review of selected academic papers, industry reports, and related articles. The main set of papers used for this project focuses on multilingual voice interaction, AI (especially speech / language tech like ASR, NLU, DM, TTS, MT, and LLMs) in cars, and cross-cultural HMI and user experience issues.

The papers were mainly gathered from peer-reviewed conference proceedings (e.g., INTERSPEECH, ACL, ICASSP, CHI, CSCW, CUI, specialized automotive HMI venues), top journals, and important pre-print archives available within the provided dataset. We prioritized studies that specifically covered:

- Multilingual features in voice systems.
- AI methods for processing spoken language in multiple languages.
- Uses or direct relevance for smart car environments.
- Cross-cultural usability, user experiences for diverse language groups, or voice interaction challenges for specific user groups.
- New AI models, comparisons of existing methods, or solid evaluation techniques.

Chapter 2

Literature Review

Developing good multilingual voice assistants (VAs) for smart cars relies on several Artificial Intelligence (AI) techniques working together. These techniques cover everything from understanding spoken language in different languages to giving natural-sounding replies. This chapter classifies these main AI techniques, providing an organized overview. We group them into Core Speech Processing Pipeline Components, Enabling and Supporting Technologies, and Advanced and Specialized Techniques.

2.1 Core Speech Processing Pipeline Components

The basic working of any voice assistant uses a core pipeline to handle speech input, understand its meaning, manage the conversation, and produce spoken output. In a multilingual setup, each of these components needs special AI techniques to handle linguistic diversity.

2.1.1 Multilingual Automatic Speech Recognition (ASR) / Speech-to-Text (STT)

Multilingual ASR is the basic tech that converts speech in different languages into computer-readable text. Its accuracy is crucial, as mistakes here propagate through the entire system. Key aspects include:

- **End-to-End (E2E) Models:** Modern ASR systems often use E2E neural setups, like those based on Transformers, which learn acoustic and language parts together. Transducer models (e.g., LAMASSU [1]) excel for streaming ASR and speech translation, vital for quick in-car responses. Attention-based sequence-to-sequence models are also common, forming the base of many general-purpose speech recognition systems adapted for multilingual use [25, 26]. These E2E methods differ from older hybrid systems (e.g., HMM-DNN) by directly mapping speech to text units, potentially being better for handling multiple languages.

- **Techniques for Handling Accents, Dialects, and Noise:** A major challenge for in-vehicle ASR is dealing with diverse accents, dialects, and cabin noise (engine, road, AC, passenger talk) [4, 5]. AI solutions include data augmentation, multi-task learning, and adaptive techniques for specific speakers or accents. Specialized datasets reflecting these variations (e.g., [22]) are key. Some systems, like Google Cloud API solutions, inherently handle some of these factors [17].

- **Language Identification (LID):** In multilingual VAs, crucial to correctly identify the spoken language(s). LID can be a preprocessing step for language-specific ASR or integrated into the ASR model for seamless language switching or handling code-switched input [18]. Language-agnostic ASR models (e.g., LAMASSU [1]) aim to process speech without prior LID by learning shared cross-lingual features. AI4Bharat's IndicLID exemplifies focused LID for related languages [18].

2.1.2 Multilingual Natural Language Understanding (NLU)

Once speech is text, Multilingual NLU techniques extract meaning, user intent, and relevant entities (slots) from the text, irrespective of language or structure. Key tasks include:

- **Intent Recognition Across Languages:** Classifying the user's goal (e.g., "navigate to," "play music") from text in various languages, often using multilingual classifiers or cross-lingual transfer learning [19, 20].
- **Slot Filling and Entity Extraction in Multilingual Contexts:** Identifying specific parameters for an intent (e.g., "Airport" as destination). These methods use multilingual pre-trained models and transfer learning for language-specific entity variations [2, 19, 27].
- **Cross-lingual Transfer Learning for NLU Models:** A dominant approach uses large pre-trained language models (e.g., mBERT, XLM-R) trained on multilingual text [27]. These models learn cross-lingual alignments, allowing knowledge transfer from high-resource to low-resource languages, enabling NLU with limited target-language data [9, 13]. The work by Gerz et al. [20] systematically studies these methods, using models like mUSE and LaBSE.

2.1.3 Dialogue Management (DM) for Multilingual Interactions

Dialogue Management (DM) tracks conversations, decides system actions, and controls multi-turn interaction flow. For multilingual VAs, DM faces added complexities like:

- **State Tracking in Mixed-Language Dialogues:** Tracking conversational state even with language switches or code-mixing, requiring language-agnostic semantic representations [13, 14].
- **Strategies for Clarification and Disambiguation:** If ASR / NLU uncertainty arises (especially multilingually due to accents, code-switching, low-resource languages), DM needs robust, culturally appropriate clarification strategies [21, 28].
- **Contextual Carry-over Across Languages:** Maintaining context (e.g., previous entities / intents) if the interaction language changes is a key multilingual DM challenge.

While not always the primary focus of the reviewed papers, robust DM is crucial for effective multilingual ASR and NLU within a VA [28].

2.1.4 Multilingual Speech Synthesis (TTS) / Natural Language Generation (NLG)

After the VA decides on a reply, Multilingual TTS and NLG convert this idea into natural-sounding speech in the target language(s). Core aspects include:

- **Generating Natural-sounding Speech in Multiple Languages and Voices:** Modern neural TTS systems (e.g., Tacotron, FastSpeech) aim for intelligible and natural-sounding speech (prosody, fluency) across many languages [12, 29].
- **Voice Cloning and Cross-Lingual Voice Transfer:** Synthesizing speech in a target language using a source language speaker's voice characteristics, or ensuring a consistent VA voice across languages [18]. StyleS2ST focuses on zero-shot style transfer for direct speech-to-speech translation [26].
- **Prosody and Emotion Modeling in Multilingual TTS:** For engaging interactions, especially in cars, TTS systems are being developed to generate speech with appropriate emotional coloring and prosody across languages [3, 30].

2.2 Other Helpful AI Technologies

Besides core pipeline components, other AI technologies enhance multilingual VAs in cars.

2.2.1 Machine Translation (MT)

Machine Translation converts text from one language to another. Key considerations:

- **Neural Machine Translation (NMT):** Transformer-based NMT models are current state-of-the-art. For VAs, NMT can be used in cascaded systems (e.g., ASR Lang A → MT to Lang B → NLU Lang B [2]) or for data augmentation (e.g., delexicalized translation for NLU training [19]).
- **Real-time Translation for Quick Responses:** In-car interactions demand speed, so VAs need fast MT for near-instant user input / system reply translation [9, 13]. StyleS2ST targets direct speech-to-speech translation to reduce ASR-MT-TTS latency [26].

2.2.2 Large Language Models (LLMs)

LLMs are powerful tools for enhancing multilingual VAs due to their extensive world knowledge and strong multi-language understanding / generation. Key aspects include:

- **Pre-trained Multilingual LLMs:** Models like BLOOM, PaLM, and regional models (e.g., AI4Bharat's MuRIL [18]) trained on massive multilingual corpora provide strong foundations, often with impressive zero / few-shot capabilities for new languages [9].
- **Fine-tuning LLMs for In-Vehicle Tasks:** General LLMs can be fine-tuned on domain-specific datasets (e.g., in-vehicle commands) to adapt their knowledge and response style for the automotive context (e.g., for CAV driving assistance [8, 15]).
- **LLMs for Rapid Language Expansion:** LLMs' inherent multilingual capabilities allow quick VA support for new languages with minimal / no language-specific data, vital for low-resource languages.

Systems like Talk2Care [15] and AISA [31] demonstrate LLM use in specialized conversational contexts, showing potential for sophisticated dialogue understanding but with challenges in real-time performance and in-vehicle resource requirements [8].

2.2.3 Knowledge Bases and Information Retrieval

To give useful driving-related answers (POIs, traffic, diagnostics), VAs need access to diverse, possibly multilingual, knowledge sources. AI is used for:

- **Querying Multilingual Knowledge Graphs / Databases:** Understanding queries in one language to retrieve info stored in another.
- **Cross-lingual Information Retrieval (CLIR):** Finding relevant data across language barriers.
- **Multilingual Information Presentation:** Organizing found information so it makes sense when presented in the user's preferred language.

While not direct language processing, effective IR is vital for making the VA useful.

2.3 Advanced and Specialized AI Techniques

Beyond the core pipeline and general enabling technologies, special AI methods are needed to handle more complex language situations and user needs in smart cars.

2.3.1 Code-Switching Detection and Handling

Code-switching, switching languages mid-conversation or mid-sentence, is common for multilingual speakers. AI is needed to:

- **Identify Code-Switches:** Detect when and how languages are mixed, which can follow complex linguistic patterns [11, 14].
- **Robust Models for Code-Switched Input:** Standard ASR / NLU often fail with code-switched input. Specialized models using language-agnostic representations or explicit switch-point modeling are being developed [10, 21].
- **Generate Code-Switched Responses:** Some users may prefer VAs that also generate code-switched replies for a more natural interaction [14].

2.3.2 Emotion-Aware Interaction Models

The driver's emotional state is crucial for safety and experience. Emotion-aware VAs aim to:

- **Detect Driver Emotion:** Analyze speech acoustics, linguistic content, and potentially other sensor data (e.g., in-cabin cameras, wearables) to infer driver emotions like stress or fatigue [3, 30].
- **Adapt VA Responses to Emotion:** Tailor replies and interaction style (e.g., empathetic tone) or suggest actions (e.g., a break if fatigue is detected) to enhance safety and comfort [5].

2.3.3 Personalization and User Modeling

Tailoring VA behavior to individual users is key for engagement and usability in multilingual settings, involving:

- **Adapting to Individual Language Profiles:** Learning a user's languages, code-switching habits, accent, and common vocabulary over time [4, 7].
- **Learning from Past Interactions:** Using machine learning to refine understanding and response strategies from historical user interaction data.

2.3.4 Language-Agnostic and Universal Representations

A major trend in multilingual AI is developing models that learn language-agnostic or universal semantic representations:

- **Shared Semantic Spaces:** These models map words / sentences from different languages with similar meanings to nearby points in a common embedding space, facilitating cross-lingual understanding without explicit translation [1, 20] (e.g., LaBSE, mUSE).

Chapter 3

Detailed Analysis: Technologies, Features, Trends, and Comparisons

This chapter takes a closer look at the Artificial Intelligence (AI) techniques and models we classified in Chapter 2. A comparative summary of these technologies—outlining their purpose, key in-vehicle challenges, state-of-the-art examples, prevailing trends, and primary applications—is presented in Table 3.1. The following sections provide a brief narrative synthesis to contextualize this information and highlight overarching insights for each technological area. The main goal is to use this combined table-and-text approach to understand the strengths, weaknesses, and practical applicability of these methods for enabling effective and culturally sensitive voice interactions in smart cars.

Table 3.1: Comparative Summary of Key AI Techniques for In-Vehicle Multilingual Voice Assistants

AI Technique	**Brief Definition / Purpose**	**Key In-Vehicle Challenges**	**SOTA Examples & Core Approach (from cited literature)**	**Key Trends**	**Primary Applications**
Multilingual ASR / STT	Converts speech from multiple languages into text, forming the VA's input.	Low-resource language accuracy; In-vehicle noise robustness; Accent / dialect handling; Low latency; Accurate Language ID.	OpenAI Whisper (Large pre-trained); LAMASSU [1] (Streaming E2E Transducer); IndicWav2Vec [18] (SSL).	Shift to E2E neural models; Rise of SSL; Focus on language-agnostic representations.	Command & Control; Dictation; Information Search input.
Multilingual NLU	Extracts meaning, user intent, and entities / slots from multilingual text.	Intent / slot accuracy across languages; ASR error robustness; Multi-turn context; Ambiguity resolution.	PLMs (mBERT, XLM-R [13, 27], IndicBERT [18]); LLMs (GPT series [8]); Sentence Encoders (LaBSE, mUSE [20]); Zero-shot algorithms [19].	Transformer PLMs; LLMs for NLU; Zero / few-shot learning.	Understanding commands for navigation, infotainment, vehicle control.

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AI Technique	Brief Definition / Purpose	Key In-Vehicle Challenges	SOTA Examples & Core Approach (from cited literature)	Key Trends	Primary Applications
Multilingual DM	Manages conversation flow, tracks state, decides system actions.	Mixed-language state tracking; Clarification / disambiguation strategies; Cross-lingual context carry-over; Initiative management.	Rule-based / FSA DMs [7]; LLMs as implicit DMs [8, 15, 28].	Shift from rule-based to data-driven / hybrid; LLMs for E2E dialogue; Proactive DM.	Orchestrating multi-turn tasks; Adaptive assistance; Proactive suggestions.
Multilingual TTS / NLG	NLG formulates text response; TTS converts text to natural-sounding speech.	Naturalness / intelligibility across languages; Voice consistency; Prosody / emotion modeling; Low latency; OOV handling; Low computational footprint.	Neural TTS (Tacotron, FastSpeech based); Indic-TTS [18]; Voice cloning / style transfer techniques [26].	Dominance of neural TTS; E2E models; Expressive TTS; Lightweight models for on-device use.	Delivering VA responses; Reading messages / news; Safety alerts.
Machine Translation (MT)	Automated text conversion between languages; used in cascaded VAs or for data augmentation.	Accuracy / fluency; Low latency; Automotive domain adaptation; Handling idioms; Low-resource pairs.	NMT (IndicTrans2 [18], NLLB [2]); Direct S2ST / ST (LAMASSU [1], StyleS2ST [26]); APIs (Google Translate [32]).	Transformer NMT dominance; Large pre-trained NMTs; Research into direct S2ST / ST.	Cascaded VA pipelines; User translation features; Data augmentation.
Large Language Models (LLMs)	Massive models for advanced understanding, generation, and reasoning.	Latency; Computational cost / on-board deployment; Hallucinations / factuality; Controllability / safety; Domain adaptation; Ethical alignment.	Proprietary (GPT series [8, 15]); Open-source (LLaMA [18]); Specialized (DriveGPT4 [28]).	Rapid model size / data escalation; Multi-modality; Efficiency research; Safety / alignment focus.	Advanced dialogue; Reasoning / explanation; Personalized assistance.

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AI Technique	Brief Definition / Purpose	Key In-Vehicle Challenges	SOTA Examples & Core Approach (from cited literature)	Key Trends	Primary Applications
KB / IR for Driving Context	Accessing specific, dynamic, structured info (maps, POIs, traffic, diagnostics).	Data accuracy / timeliness; Multilingual query / presentation; Dialogue integration; Ambiguity; Offline access.	RAG with LLMs [15]; Ontologies [3, 6]; DM for info seeking [13].	Increasing LLM+KB integration (RAG); Multilingual QA; Automotive knowledge graphs.	Navigation guidance; Vehicle info / diagnostics; Local services / news.
Code-Switching Handling	Detecting and understanding mixed-language speech / text.	Accurate switch point ID; Robust ASR / NLU on mixed input; Semantic coherence; Data scarcity.	User studies on preferences [14, 21]; Language-agnostic models (e.g., LAMASSU [1]).	Growing demand for CS support; E2E models for robustness; User adaptation.	Natural interaction for multilinguals; Reduced user frustration.
Emotion-Aware Interaction	Recognizing and responding to user emotions for safer, empathetic interaction.	Multimodal complexity; Contextual understanding of emotion; Cultural variations; Ethics / privacy.	Conceptual frameworks [3]; SER models using multimodal features [30].	Shift to multimodal sensing; Deep learning for emotion; Focus on ethics / privacy.	Driver state monitoring; Empathetic interaction; Comfort / well-being.
Personalization	Adapting VA behavior and style to individual users multilingually.	Adapting to linguistic profile (accent, vocab, CS); Learning habits; Privacy; Cold start; Multi-user handling.	Adaptive DM [13]; LLM contextual history [15]; User-initiated customization [33].	Dynamic systems that learn; On-device / federated learning for privacy; User control over data.	Tailored VA experience; Reduced friction; Proactive assistance.

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AI Technique	Brief Definition / Purpose	Key In-Vehicle Challenges	SOTA Examples & Core Approach (from cited literature)	Key Trends	Primary Applications
Language-Agnostic Representations	Models learning language-independent semantic / acoustic representations.	True universality across diverse languages; "Curse of multilinguality"; Interpretability; Retaining nuances.	mPLMs (mBERT [27]); Sentence Embedders (LaBSE, mUSE [20]); ASR (LAMASSU [1], Whisper [9]).	Push for universal representations; SSL for learning; Balancing universality with language-specificity.	Scalability; Handling unforeseen languages; Better CS support.
Multimodal Sensor Fusion	Combining data from diverse vehicle sensors for holistic context.	Disambiguating voice; Rich driver state modeling; Sensor sync / load; Robust fusion algorithms.	Conceptual need [3]; Implied by ADS perception [6].	Deeper integration of voice with visual / other sensory data for context-aware VAs.	Improved safety and trust through better VA context awareness.

3.1 Multilingual Automatic Speech Recognition (ASR) / Speech-to-Text (STT)

3.1.1 Description

Multilingual Automatic Speech Recognition (ASR) is the foundational technology that converts human speech from multiple languages into written text. Its accuracy is paramount, as errors propagate through the entire VA pipeline, affecting subsequent Natural Language Understanding (NLU) and Dialogue Management (DM) stages [13]. In-vehicle ASR must perform robustly across a wide array of languages, accents, and challenging acoustic environments.

3.1.2 Key Features & Challenges for In-Vehicle VAs

The automotive cabin presents a unique set of challenges for ASR. A primary hurdle is achieving high accuracy for low-resource languages, which lack the extensive speech corpora needed to train robust models, a problem highlighted in studies on Welsh and Indian languages [12, 18]. Another major challenge is robustness to significant in-vehicle noise from the engine, road, and passengers [7]. Furthermore, effectively handling diverse user accents and dialects remains a persistent problem and a primary source of user frustration [4, 5, 10, 11]. Finally, the system must perform this complex processing with minimal latency for safe, real-time interaction, often on computationally constrained in-vehicle hardware [1, 22].

3.1.3 State-of-the-Art Models and Methods

Current multilingual ASR relies heavily on neural networks. End-to-End (E2E) models are a dominant trend, including streaming Transducer models like LAMASSU, which are well-suited for low-latency in-vehicle tasks [1]. Another key development is large pre-trained models like OpenAI's Whisper, which demonstrate impressive zero-shot performance and broad language coverage [9, 18]. To address data scarcity for low-resource languages, Self-supervised Learning (SSL) from unlabeled audio, as exemplified by IndicWav2Vec for Indian languages, has become a crucial technique [18]. The literature also explores direct E2E speech translation (ST) models, which, by avoiding a separate ASR step, can mitigate error propagation from cascaded ASR-MT systems [2, 26].

3.1.4 Trends and Comparative Analysis

The ASR field has seen a definitive shift from older statistical models (HMM-GMMs) to E2E neural architectures, which simplify the pipeline. This shift has enabled key trends like SSL, reducing dependency on labeled data, and the pursuit of language-agnostic models that process multiple languages without explicit identification [1]. A central trade-off exists between low-latency streaming models optimized for speed and large offline models optimized for accuracy [1]. Similarly, there is a distinction between the broad, general-purpose coverage of models like Whisper and the deep, targeted coverage of regional initiatives like AI4Bharat, a choice often dictated by market needs [9, 18].

3.1.5 Application in Intelligent Vehicles

Multilingual ASR is fundamental for core in-vehicle VA functions. This includes command and control of vehicle systems (climate, infotainment, navigation), hands-free dictation of messages or notes, and serving as the input for information retrieval queries [7, 22, 23].

3.2 Multilingual Natural Language Understanding (NLU)

3.2.1 Description

Following ASR, Multilingual NLU computationally analyzes the transcribed text to extract its meaning, discern user intent, and identify key entities (slots). Robust multilingual NLU is essential for the VA to respond appropriately to driver requests across a spectrum of languages.

3.2.2 Key Features & Challenges

Effective multilingual NLU must reliably perform intent recognition (identifying the user's goal) and slot filling (extracting parameters like destination or song title) across different linguistic structures [2, 19, 34]. A major challenge is its dependency on cross-lingual transfer learning to bootstrap capabilities in low-resource languages by leveraging data from high-resource ones [20, 27]. Furthermore, NLU systems in cars must be robust to ASR transcription errors and understand conversational context and dialogue history to resolve ambiguities in multi-turn interactions [13, 28].

3.2.3 State-of-the-Art Models and Methods

Modern multilingual NLU is dominated by Transformer-based Pre-trained Language Models (PLMs). Models like mBERT and XLM-R are pre-trained on vast multilingual corpora and can be fine-tuned for specific intent and slot-filling tasks, a process detailed in studies like [27] and [13]. Region-specific models like IndicBERT further show the value of targeted pre-training for language families [18]. Increasingly, Large Language Models (LLMs) are used directly for NLU, often performing well in zero-shot or few-shot settings [8]. Other specialized approaches include training on translated data and using sentence encoders like LaBSE and mUSE for robust cross-lingual intent detection [20].

3.3 Dialogue Management (DM) for Multilingual Interactions

3.3.1 Description

Dialogue Management (DM) is the VA's cognitive core, orchestrating the interaction. It interprets NLU outputs, maintains conversational state across multiple languages and turns, decides on system actions (e.g., controlling vehicle functions), and plans responses to ensure coherent, context-aware conversations [13].

3.3.2 Key Features & Challenges

The in-vehicle context presents significant DM challenges. The system must track dialogue state through mixed-language conversations (code-switching) [14, 21]. It requires robust, culturally appropriate strategies for asking clarifying questions when faced with ambiguity or errors [5, 28]. Maintaining conversational context across language switches is another key difficulty. Furthermore, the DM must manage interaction initiative, balancing proactive suggestions based on vehicle or driver state with direct responses, a task complicated by varying cultural expectations [4, 21].

3.3.3 State-of-the-Art Models and Approaches

While still evolving for the multilingual vehicle domain, relevant DM approaches include modular pipelines, simpler rule-based or Finite-State Automaton (FSA) DMs for basic tasks [7], and data-driven methods using statistical or reinforcement learning (RL) techniques. A major emerging trend is the use of Large Language Models (LLMs) as implicit dialogue managers. By conditioning on conversational history, LLMs can handle state tracking and response generation implicitly, effectively subsuming traditional DM roles [8, 15], though with challenges in controllability and safety for critical scenarios.

3.4 Multilingual Speech Synthesis (TTS) / Natural Language Generation (NLG)

3.4.1 Description

Once a response is decided, Natural Language Generation (NLG) formulates it as coherent text, and Text-to-Speech (TTS) converts this text into audible speech. The quality of this output significantly impacts the perceived naturalness and intelligence of the VA.

3.4.2 Key Features & Challenges

A primary goal is generating clear, natural-sounding speech across diverse languages, which involves mastering each language's unique prosody and phonetic characteristics [13, 29]. Maintaining a consistent voice persona across languages is crucial for user experience but technically challenging [3, 35]. Other key challenges include modeling emotion and prosody for more engaging interactions [30], handling out-of-vocabulary (OOV) terms, and achieving low latency and a small computational footprint for on-board deployment.

3.4.3 State-of-the-Art Models and Approaches

Modern TTS has shifted to neural network-based synthesis, such as Tacotron or FastSpeech, which offer high naturalness. This includes regional models like Indic-TTS from AI4Bharat [18]. The field has also seen advances in voice cloning and style transfer techniques, allowing for a consistent voice across languages or expressive emotional tones [26]. The dominant trend is toward End-to-End models that directly synthesize speech from text, simplifying the pipeline and improving quality.

3.5 Enabling and Advanced Technologies

Beyond the core pipeline, several other AI technologies augment VA functionality and address more nuanced aspects of interaction.

3.5.1 Enabling Technologies: MT, LLMs, and KB/IR

Machine Translation (MT) serves a dual role, either as a component in a cascaded system or as a tool for creating synthetic training data to support other models [2, 19]. Large Language Models (LLMs) are transformative, showing impressive zero-shot capabilities and advanced reasoning, but their practical deployment is currently hindered by latency, cost, and safety concerns [8]. Knowledge Base and Information Retrieval (KB/IR) systems ground VA responses in factual, domain-specific data, such as map or diagnostic information, with Retrieval-Augmented Generation (RAG) being a key trend for integrating LLMs with external knowledge [6, 15].

3.5.2 Advanced and Specialized Techniques

To create a truly user-centric experience, VAs must handle more complex user behaviors and contexts. A primary challenge is handling **Code-Switching**, the alternation between languages, to support the natural speech patterns of multilingual users [14, 21]. Another key area is **Emotion-Aware Interaction**, which leverages multimodal sensor data to make the VA more empathetic and responsive to driver state, enhancing safety and satisfaction [3, 30]. Central to these advancements is **Personalization and User Modeling**, which enables the VA to learn and adapt to an individual's accent, vocabulary, and preferences over time, though this introduces significant privacy considerations [4, 7]. Underpinning many of these techniques is the development of **Language-Agnostic Representations** for scalable and efficient multilingual processing [1, 20].

Chapter 4

Cross-Cultural and User-Specific Challenges

While the AI techniques discussed in Chapter 3 show significant progress, creating universally effective voice assistants (VAs) remains a profound challenge. The immense diversity of human language, cultural norms, and individual user needs presents hurdles that current technology does not fully accommodate. This chapter examines these key issues—linguistic variation, cultural appropriateness, and specific user group needs—which must be addressed for the widespread and ethical adoption of multilingual VAs in the global automotive market.

4.1 Linguistic Variation and Robustness

Language is not monolithic; its variations pose significant problems for ASR and NLU systems.

4.1.1 Accents and Dialects

A persistent, widely reported failure of VAs is their inability to reliably understand diverse accents and regional dialects [4, 5, 10, 11, 21, 32]. Standard ASR models, often trained on "standard" accents, exhibit higher error rates for non-standard or non-native speakers. User-centric studies consistently highlight this problem: frustrations among bilingual users [11, 21], feelings of exclusion among multicultural users [5], and struggles for speakers of regional dialects, such as Chinese older adults, are well-documented [4].

The root cause is a lack of training data representing this phonetic and prosodic diversity. Consequently, there is a critical need for adaptive ASR models that can either be fine-tuned for specific accents or learn a user's accent over time through continued interaction.

4.1.2 Low-Resource Languages

Many of the world's over 7,000 languages are "low-resource" for AI development, lacking the large datasets needed to train robust models from scratch [9]. This digital inequality is a major barrier to creating truly global products. For instance, initiatives like AI4Bharat are required to build foundational models for the many languages of India [18].

To address this, methods like **cross-lingual transfer learning, zero-shot learning, and few-shot learning** are crucial [19, 20, 27]. These techniques leverage knowledge from data-rich languages to bootstrap capabilities in data-scarce ones, such as by fine-tuning multilingual models like mBERT or using universal ASR models like Whisper [9]. However, performance in these low-resource scenarios often remains significantly below that of well-supported languages, especially for complex in-vehicle commands.

4.1.3 Code-Switching

Code-switching—alternating between languages mid-conversation—is a natural communication pattern for multilingual communities worldwide [11, 14]. Current VAs, which typically operate in a monolingual mode, are ill-equipped to handle this behavior. This creates technical challenges for ASR, NLU, and DM, as they struggle to transcribe, understand, and track state in mixed-language contexts. Research consistently shows a strong user preference for VAs that can understand code-switching, as its absence leads to user frustration and makes interaction feel unnatural [14, 21]. Developing robust models to process mixed language is a key research area for creating truly user-centric VAs.

4.2 Cultural Nuances and Contextual Appropriateness

Effective VA communication requires not just linguistic accuracy but also respect for cultural nuances and social norms. Failures can lead to misinterpretation, user dissatisfaction, and the perception of the VA as culturally inept.

4.2.1 Culturally Embedded Expressions and Social Norms

Languages are rich with culturally specific idioms, metaphors, and norms of politeness and formality that do not translate literally. VAs trained on formal text often fail to understand these subtleties, leading to communication breakdowns and a failure to meet user expectations [4, 21]. Therefore, multilingual VAs require more **culturally-aware NLU and NLG**—trained on more diverse data and capable of adapting language style (e.g., formality) to fit cultural context and user preference [3].

4.2.2 Impact on User Trust, Acceptance, and Safety

These failures in cultural awareness can severely erode user trust and hinder acceptance [3, 5]. A VA perceived as culturally insensitive or "clueless" is unlikely to be trusted for more than basic tasks. For multicultural users in particular, such misrecognitions can feel alienating [5]. Critically, in an automotive context, misunderstandings from linguistic or cultural gaps can have direct safety implications. A VA misinterpreting an urgent but colloquial command could fail to perform a necessary safety-related action, making culturally competent interaction essential for operational safety.

4.3 Specific User Group Needs

Beyond general diversity, specific user groups have distinct challenges and expectations when interacting with in-vehicle VAs. Addressing these is essential for inclusive HMI design.

4.3.1 Older Adults

Older adults, a growing demographic of drivers, often face unique challenges with VAs. These include higher barriers to technology adoption; cognitive and sensory considerations requiring simpler interfaces and clearer, slower feedback [15]; and specific speech patterns that can challenge standard ASR models. Their motivations may also extend beyond utility to include social companionship [4]. Furthermore, as studies with Chinese older adults show, a lack of support for regional dialects and local cultural references is a major source of frustration and abandonment [4].

4.3.2 Non-Native Speakers / Second Language (L2) Users

L2 users represent a vast global user base and often experience a higher cognitive load when interacting with VAs, especially under driving stress [11]. Their accents are a major hurdle for ASR, and they may be more sensitive to recognition errors [31]. The natural tendency of L2 users to code-switch is another

key challenge for current systems [21]. As research by Cihan et al. suggests, designing interfaces that are "bilingual by default" and supportive of these interaction styles is crucial [11].

4.3.3 Multicultural and Other User Groups

For multicultural users, VA interactions intersect with their sense of identity. Misrecognitions related to accents or culturally specific names can be perceived as alienating "microaggressions," whereas culturally sensitive interactions can be affirming [5]. These users expect a degree of cultural competence that current VAs often lack [21]. Similarly, users with speech impairments face extremely high ASR error rates, rendering VAs largely unusable without specialized models and deep personalization, a known accessibility challenge in AI.

4.4 Personalization and Adaptive Interfaces

Addressing the diverse needs outlined above requires a strong focus on **personalization and adaptive interfaces**. A one-size-fits-all approach is insufficient for a global user base. VAs must adapt to an individual user's linguistic profile, cultural background, and interaction preferences over time [3, 7]. This involves more than just selecting a language; it means dynamic language switching [33], accent adaptation, accommodation of code-switching styles [14], learning user-specific vocabulary, and deploying adaptive error-handling strategies. The overarching goal is a VA that feels tailored to the individual, fostering trust and ensuring the assistant is a helpful co-pilot rather than a source of frustration [4, 5]. This requires advanced user modeling and a commitment to designing with diverse users at the forefront.

Chapter 5

Evaluation and Critical Assessment of Multilingual VAs in Vehicles

Developing effective and reliable multilingual voice assistants (VAs) for smart cars requires careful evaluation. This chapter describes comprehensive criteria for judging these systems, synthesizes findings from the literature to test several key hypotheses, and critically assesses the practical and ethical implications of current applications. The goal is to evaluate not just technical performance, but also the suitability, safety, and user-friendliness of these VAs for a diverse global user base.

5.1 Establishing Comprehensive Evaluation Criteria

A holistic evaluation of multilingual in-car VAs must extend beyond component-level accuracy scores to encompass a range of technical, system-level, and user-centered metrics. Based on the reviewed literature, the following criteria are essential.

5.1.1 Component-Level Performance Metrics

The foundational performance of each AI component must be measured.

- **ASR Performance:** Key metrics include Word/Character Error Rate (WER/CER) for transcription accuracy [1, 12, 22], Language Identification (LID) accuracy [18], and specific testing on diverse accents and in noisy in-vehicle conditions [5].
- **NLU Performance:** Metrics include Intent Recognition Accuracy, Slot Filling F1-score [19, 27], and the stricter Frame Accuracy (correct intent and all slots) [20].
- **MT Performance:** For VAs using translation, quality is assessed with metrics like the BLEU score [2, 17] and complementary scores such as METEOR or TER.
- **TTS Performance:** Quality is judged by the subjective Mean Opinion Score (MOS) for naturalness and intelligibility [18, 30], as well as speaker similarity for persona consistency and intelligibility in noise.

5.1.2 System-Level and User-Centered Evaluation

Beyond individual components, the VA must be evaluated as a complete system from the user's perspective.

- **End-to-End Task Completion Rate:** The ultimate measure of effectiveness—whether the VA can successfully help a user complete a full task (e.g., "Navigate to the nearest restaurant and call them"). This reflects the integrated performance of the entire pipeline.

- **Latency and Real-Time Responsiveness:** The end-to-end delay from user utterance to VA response is critical for safety and natural interaction. This has been a key factor in evaluating both streaming ASR systems [1] and LLMs [8].

- **Code-Switching Handling:** Evaluation requires specific metrics for mixed-language input, such as code-switch point detection accuracy and performance of ASR/NLU on such utterances [14, 21].

- **User-Centered Metrics:** Measuring the user experience is paramount. This includes perceived usability via scales like SUS [15]; subjective ratings of satisfaction, trust, and cognitive load [11, 31]; qualitative assessments of cultural appropriateness and inclusivity [4, 21]; and impact on driver safety (e.g., eyes-off-road time).

- **Robustness and Generalization:** The VA must be tested on unseen languages, dialects, and accents, as well as in novel acoustic environments, to assess how well its performance generalizes beyond its training data [34].

5.2 Testing Hypotheses by Comparing Existing Work

By synthesizing the surveyed research, we can evaluate several key hypotheses regarding different AI approaches for multilingual in-vehicle VAs.

- **Hypothesis 1: E2E multilingual ASR/ST models offer superior latency for in-vehicle VAs compared to cascaded ASR+MT systems.** *Evidence Summary:* Research on streaming E2E architectures like LAMASSU emphasizes inherent low-latency benefits [1]. Studies on direct speech translation also suggest that E2E models can reduce error propagation from cascaded pipelines, making them well-suited for the real-time demands of in-car interaction [2].

- **Hypothesis 2: Fine-tuning large, pre-trained multilingual models is the most effective practical strategy for developing in-vehicle NLU/ASR capabilities across multiple languages.** *Evidence Summary:* Building models from scratch per language is resource-prohibitive. The literature consistently shows that fine-tuning large models like mBERT or Whisper provides a strong balance of performance and development effort. While zero-shot transfer offers a baseline, fine-tuning, even with limited data, typically yields superior, more adapted performance for specific languages and domains [18, 20, 27].

- **Hypothesis 3: User acceptance and trust are more significantly impacted by a VA's ability to handle real-world linguistic variations (accents, code-switching) than by its raw accuracy on standard, clean test sets.** *Evidence Summary:* There is a strong consensus from user-centric studies. Research repeatedly documents that user frustration, alienation, and lack of trust stem from failures to understand diverse accents or accommodate natural code-switching, often outweighing marginal gains on standardized benchmarks that lack real-world diversity [4, 5, 10, 14, 21].

- **Hypothesis 4: While LLMs show high potential for complex reasoning in vehicles, their practical deployment is currently hindered by latency, cost, and safety concerns.** *Evidence Summary:* Studies demonstrate that advanced LLMs possess impressive domain knowledge and dialogue capabilities [8, 15, 28]. However, the same research quantifies the significant challenges of high latency, API costs, and the risks of factual incorrectness ("hallucinations"), which are critical barriers to their safe, reliable, and cost-effective deployment in real-time in-vehicle systems [8].

5.3 Evaluating Practical Relevance and Critically Assessing Applications

A critical assessment must consider the practical, real-world applicability of these AI techniques beyond laboratory benchmarks.

5.3.1 Bridging the Gap Between Lab Performance and Real-World Utility

A persistent challenge is the discrepancy between performance in controlled lab settings and actual utility in the dynamic, noisy in-vehicle environment. Standard test sets may not adequately capture the true range of acoustic conditions, diverse accents, or naturalistic speech patterns like code-switching [10, 21]. Evaluating practical relevance demands more real-world testing and a focus on user-perceived quality over purely automated metrics.

5.3.2 Safety, Scalability, and Ethics in the Automotive Context

The in-vehicle environment is inherently safety-critical, where a VA error can cause driver distraction. Current evaluation methods often lack robust frameworks for assessing safety and reliability, especially for LLMs [3, 8]. Furthermore, supporting a global market presents immense scalability and maintenance challenges, requiring significant resources to ensure quality for each language variant [18, 34]. This raises ethical considerations that must be addressed:

- **Algorithmic Bias and Inclusivity:** Ensuring models perform equitably across all linguistic groups and do not perpetuate digital inequality or cause user harm is paramount [5, 9].
- **Data Privacy:** As VAs can capture sensitive data, robust privacy-preserving techniques and transparent user-controlled policies are non-negotiable [3, 4].
- **Explainability and Trust:** In safety-critical situations, users need some level of understanding for unexpected VA behaviors. Opaque "black-box" models can hinder the trust necessary for user acceptance.

Chapter 6

Conclusion

This research summary has systematically reviewed current Artificial Intelligence (AI) techniques for multilingual voice assistants (VAs) in intelligent vehicles. We have surveyed the core processing pipeline (ASR, NLU, DM, TTS), enabling technologies like MT and LLMs, and advanced methods for handling challenges such as code-switching and emotion awareness. Throughout this paper, a central focus has been maintained on the significant hurdles posed by linguistic and cultural diversity and the imperative to create VAs that are user-focused, inclusive, and safe for a global audience.

6.1 Summary of Key Findings and Dominant Trends

Our literature review reveals several key findings and trends:

- **Dominance of E2E and Pre-trained Models:** The field has shifted to End-to-End (E2E) neural architectures [1, 2] and relies heavily on large pre-trained multilingual models like mBERT for NLU and Whisper for ASR. These facilitate cross-lingual transfer learning, which is essential for supporting low-resource languages and improving development speed [8, 9, 27].
- **Growing Focus on User-Centric Challenges:** There is increasing recognition that technical accuracy alone is insufficient. Successfully handling real-world linguistic diversity—including accents, dialects, and code-switching—is crucial for user acceptance and trust [5, 10, 21].
- **Emergence of Advanced and Adaptive Techniques:** Specialized methods for emotion awareness [3, 30] and personalization [15, 31] are being explored to create more human-like in-car experiences. However, advanced models like LLMs face practical deployment hurdles, including latency, cost, and safety concerns [8].

6.2 Synthesis of Strengths, Weaknesses, and Research Gaps

Current AI approaches present a mixed landscape.

- **Strengths:** Notable strengths include improved accuracy in major languages, the viability of cross-lingual transfer learning [18, 27], and the advanced contextual understanding offered by LLMs [8]. There is also a greater research focus on the user experience [4, 5].
- **Weaknesses:** Significant weaknesses persist, including performance gaps for low-resource languages and diverse accents [10], poor handling of natural phenomena like code-switching [14], and the high computational cost of SOTA models [8]. A lack of cultural adaptation and standardized real-world evaluation benchmarks remain critical issues.
- **Identified Gaps:** This survey identifies several crucial research gaps. There is a pressing need for: 1) diverse, large-scale, in-vehicle multilingual speech corpora [34]; 2) efficient on-device AI

models suitable for vehicle hardware; 3) holistic research into proactive and culturally adaptive dialogue management; 4) realistic evaluation methodologies that incorporate safety and cross-cultural usability [6]; and 5) principled frameworks for ethical AI and harm reduction in VAs [3, 5].

6.3 Recommendations and Concluding Vision

To advance the field, we recommend a focus on several key areas. The industry should **invest in diverse data creation** [9], **prioritize AI innovation for on-device efficiency and robustness**, and **embrace user-centered and culturally-aware design** involving diverse user groups [21]. Furthermore, developers must **adhere to "Safety and Ethics by Design" frameworks** and work towards **standardized, holistic evaluation protocols** that measure real-world performance.

The journey towards creating truly seamless and universally accessible multilingual VAs requires a concerted, interdisciplinary effort. Future in-vehicle assistants must evolve from functional tools into reliable, trustworthy, and culturally sensitive co-pilots. By prioritizing inclusivity, addressing cross-cultural challenges, and placing user safety at the forefront of design, AI can be harnessed to create in-car assistants that not only enhance the driving experience but also foster a more connected and equitable global society. The ultimate vision is an intelligent vehicle that speaks every user's language, understands their unique context, and acts as a truly intelligent companion on the road.

Appendix A

Methodology for Traceability of Literature Review

This research summary is based on a systematic review and summary of a chosen set of academic papers, industry articles, technical reports, and other relevant writings. These materials specifically cover multilingual voice assistants (VAs), how Artificial Intelligence (AI) techniques in speech and language processing are used for these systems, their use in smart cars, and the related cross-cultural usability challenges. The main set of papers for this review included 35 key documents provided by the author of this summary, representing a recent look at the field up to early 2025.

A.1 Scope and Selection Criteria

Choosing and including information from these sources was guided by the following main criteria to make sure the survey was relevant and thorough for its goals:

- Multilingual features in voice systems.
- AI methods for processing spoken language in multiple languages.
- Uses or direct relevance for smart car environments.
- Cross-cultural usability, diverse language group experiences, or voice interaction challenges for specific user groups.
- New AI models, comparative analyses, or solid evaluation techniques.

The review mostly focused on peer-reviewed conference papers (e.g., INTERSPEECH, ACL, ICASSP, CHI, CSCW, CUI, specialized automotive HMI venues), top journals in AI, HCI, and car technology, as well as important pre-print archives available within the provided dataset.

A.2 Process of Review and Synthesis

Building this research summary involved these steps:

1. **Initial Check and Familiarization:** All provided documents (mostly OCR'd PDF outputs) were first reviewed to confirm they directly related to the main themes of multilingual in-car VAs and to get a general idea of the topics covered.
2. **Extracting and Grouping Information:** Key information about specific AI techniques, algorithms, models, system designs, noted features, new trends, practical challenges, user study results, and evaluation metrics was systematically pulled from each relevant paper. This extracted info was then grouped by theme according to this summary's structure (as outlined in Chapters 2 and 3).

3. **Putting Together and Comparing Information:** Information from multiple sources on each specific theme or AI technique was combined. This meant finding similarities and differences in methods (e.g., single vs. multilingual training, zero-shot vs. fine-tuning for LLMs), comparing reported performance details (e.g., accuracy, speed, language coverage of different ASR systems), and discussing different views on remaining challenges and possible solutions. Tables and descriptions in the text were planned to show these connections.

4. **Critical Check and Evaluation of Practical Relevance:** The findings and claims in the reviewed papers were critically checked for how well they apply to real-world multilingual in-car VA use. This involved considering things like whether they are computationally doable, can scale up, safety effects, ethical issues, and reported user experiences.

5. **Forming and Testing Hypotheses (Deductive):** Based on the overall ideas and patterns seen in the literature, general hypotheses (as presented in Section 5.2) were created. These hypotheses were then "tested" by gathering and cross-referencing supporting or opposing evidence found in different studies in the set.

6. **Identifying Gaps and Future Directions:** Areas where knowledge is missing, where big challenges remain, or where more research is badly needed were found based on the full review. This led to the recommendations given in the Conclusion (Chapter 6).

It's important to say again that this research summary did not involve collecting or analyzing new primary data. All conclusions, comparisons, and critical checks are strictly based on organizing, combining, and interpreting existing knowledge from the provided set of literature. The main goal was to create a structured, traceable, and full overview that critically evaluates the current state of AI techniques and cross-cultural challenges for multilingual voice assistants in smart cars. This aims to be a valuable resource for students, researchers, and people working in the field.

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